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InterpretCognates: An Interactive Toolkit and 135-Language Probing Suite for Multilingual Representation Geometry

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Summary

Massively multilingual neural machine translation (NMT) and language models map hundreds of human languages into shared high-dimensional vector spaces (NLLB Team et al., 2022; Chang et al., 2022). For researchers in psycholinguistics, bilingual cognition, and quantitative lexical typology, these internal representations offer a powerful computational testbed for investigating how lexical concepts are organized across languages, scripts, and language families (Kroll and Stewart, 1994; List et al., 2018; Jäger, 2018). However, extracting, de-confounding, and visualizing cross-linguistic representation geometry across over a hundred languages typically requires substantial custom engineering to separate surface-form and language-identity artifacts from underlying semantic structure.

`InterpretCognates` is an open-source Python and browser-based software toolkit designed to make multilingual representation probing accessible, interactive, and deterministically reproducible. Built around Meta's open-weights `NLLB-200` translation model (NLLB Team et al., 2022), `InterpretCognates` provides:

1. **A FastAPI & PyTorch Probing Backend (`backend/`)**: Extracts token- and sentence-level hidden states and cross-attention weights across any subset of NLLB-200's 200+ supported languages, supporting both contextualized carrier-frame inputs and decontextualized native bare-word tokens.
2. **A Synchronized Interactive Web Explorer (`docs/`)**: A static, zero-server browser application (hosted at `https://kylemath.github.io/InterpretCognates/`) featuring 22 interactive 2D and 3D Plotly visualizations synchronized one-to-one with our empirical study (Mathewson, 2026). Users can inspect 135-language concept manifolds, toggle mean-centering and All-but-the-Top (ABTT) anisotropy removal (Mu and Viswanath, 2018), explore phylogenetic trees against Automated Similarity Judgment Program (ASJP) phonetic distances (Wichmann et al., 2022; Jäger, 2018), and query cross-linguistic colexification networks against CLICS^3 (Rzymski et al., 2020).
3. **A Deterministic Statistical-to-LaTeX Pipeline (`paper/scripts/generate_stats.py`)**: An automated verification workflow that computes 321 statistical tests, effect sizes, null permutations, and confound controls directly from the pre-computed JSON data artifacts and compiles them into LaTeX macros (`paper/output/stats.tex`), guaranteeing zero copy-paste discrepancies between code, interactive web charts, and published prose.

Statement of Need

Researchers studying the bilingual mental lexicon (Kroll and Stewart, 1994; Kroll et al., 2010), cortical language representations (Malik-Moraleda et al., 2022; Chen et al., 2025), and cross-linguistic lexical semantics (Vulić et al., 2020) increasingly use neural language model embeddings to generate quantitative semantic norms and test cognitive hypotheses. Yet raw cosine similarities in transformer encoders are heavily confounded by representation anisotropy (Ethayarajh, 2019; Mu and Viswanath, 2018), subword tokenization fertility (Rust

et al., 2021), orthographic script sharing, and low-dimensional language-identity shifts (Chang et al., 2022; Pires et al., 2019).

Existing interpretability packages generally focus on monolingual token attribution or task-level benchmarking rather than typologically controlled psycholinguistic probing. `InterpretCognates` fills this gap by providing:

- **Pre-computed 135-Language Geometric Norms**: Ready-to-query JSON datasets covering 101 Swadesh core concepts (Swadesh, 1952), 60 non-Swadesh concrete nouns, basic color terms (Berlin and Kay, 1969), and semantic offset pairs across 135 languages (45 language families, 24 scripts) in both contextualized carrier frames and native bare-word forms.
- **Built-in Confound Controls**: Turnkey implementations of two-way Concept × Language ANOVA variance partitioning, eigenvalue Participation Ratio (PR) dimensionality estimation, ABTT principal component removal ($D = 1 \dots 10$), orthographic/phonological Levenshtein regression, and Jäger's (Jäger, 2018) off-diagonal normalized Levenshtein distance (LDND).
- **Interactive Pedagogical and Exploratory Workflows**: Psycholinguists and students can explore 3D conceptual manifolds, compare carrier-frame versus bare-word convergence, and inspect individual language families in real time in any modern web browser without requiring local GPU hardware.

Software Architecture and Reproducible Pipeline

`InterpretCognates` is organized into three modular layers (Figure 1):

1. **Extraction & Analysis Layer** (`backend/app/main.py`, `research/`): Wraps Hugging Face `transformers` (`AutoModelForSeq2SeqLM`, `AutoTokenizer`) to run multi-target translation, extract layerwise encoder hidden states (Layers $0 \dots 11$), compute attention-weighted pooled representations, and project embeddings via Principal Component Analysis (PCA) and Multidimensional Scaling (MDS).
2. **Interactive Visualization Layer** (`docs/index.html`, `docs/blog.js`, `build_static.py`): Renders pre-computed JSON artifacts (`docs/data/*.json`) into interactive Plotly charts, phylogenetic dendrograms, layerwise emergence curves, and live translation-similarity explorers. Running `python3 build_static.py` packages the entire interactive suite into a standalone static site (`docs/`).
3. **Programmatic Manuscript Synchronization** (`paper/scripts/generate_stats.py`, `paper/scripts/generate_figures.py`): Reads the exact JSON files served to the web application to generate all 16 vector PDF figures (`paper/figures/*.pdf`) and all 321 LaTeX `\newcommand` statistical macros (`paper/output/stats.tex`).

Availability

`InterpretCognates` is released under the MIT License at `https://github.com/kylemath/InterpretCognates` and archived on Zenodo at `10.5281/zenodo.23898624`, with the live interactive companion hosted at `https://kylemath.github.io/InterpretCognates/` and the companion empirical manuscript available at `arXiv:2603.02258` (Mathewson, 2026).

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Figure 1: Overview of the `InterpretCognates` interactive web explorer, 135-language geometric norm suite, and layerwise conceptual emergence visualizations.

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